DATA MANAGEMENT & SHARING PLAN

National Indigenous Health Data Pathways Infrastructure (NIHDPI)
Category I — IDSS Program

The National Indigenous Health Data Pathways Infrastructure (NIHDPI) project generates several categories of data and digital products, including operational cyberinfrastructure metadata, aggregated educational and workforce data, restricted student-level datasets maintained in federated environments, software artifacts, and public dashboards. Operational metadata will include non-identifiable logs, ETL performance metrics, and service use indicators. Aggregated programmatic data will summarize pathway participation, retention, graduation rates, and regional workforce trends, and will be fully de-identified. In contrast, identifiable or sensitive student-level information will remain under the control of the originating institutions or Tribal entities and will not be stored, hosted, or centralized by SGDI, AAIP, or IHEART. The system's federated architecture ensures that such data remain entirely within local institutional or Tribal boundaries.

NIHDPI data and metadata will follow widely accepted standards. Educational and academic data will align with the Common Education Data Standards (CEDS) and IPEDS definitions, while labor and workforce data will follow Bureau of Labor Statistics occupational classifications and HRSA shortage indicators. Metadata will use JSON-LD and schema.org conventions, accompanied by documentation describing lineage, versioning, and AI model transparency. AI model releases will include model cards and dataset documentation sheets. Indigenous Data Sovereignty frameworks—including the CARE Principles (Collective Benefit, Authority, Responsibility, Ethics) and OCAP-aligned principles (Ownership, Control, Access, Possession)—will guide all handling of Tribal-affiliated or culturally sensitive data.

Public access will be provided for aggregated datasets, analytic dashboards, system documentation, and open-source software. These materials will be disseminated through AAIP/IHEART web portals and public repositories such as GitHub. Restricted datasets—including identifiable student data, Tribal community information, and internal performance metrics—will only be accessible through institution- or Tribe-controlled federated connectors. Access will require appropriate approvals such as DUAs, IRB review (when applicable), and permissions following least-privilege principles. No identifiable data will be transferred to NSF.

Reuse and redistribution policies distinguish between open materials and protected information. Open-source software will be released under permissive licenses such as Apache 2.0 or MIT. Aggregated or synthetic datasets may be reused as permitted. Sensitive or identifiable datasets will not be redistributed, and all rights to those data remain fully with the originating institutions or Tribal nations. Any secondary use of aggregated outputs must comply with DUAs, IRB requirements, and Tribal governance expectations.

Data storage and preservation practices emphasize security and long-term accessibility of public artifacts while ensuring that sensitive data remain exclusively with their data owners. Public datasets, dashboards, and code will be archived in redundant repositories operated by AAIP and SGDI. The infrastructure will follow a zero-trust security model, incorporating role-based access control, multifactor authentication, AES-256 encryption at rest, TLS 1.3 in transit, and NIST 800-53–aligned controls. Regular penetration testing and continuous monitoring will be implemented. Identifiable data will not be stored by NIHDPI; long-term preservation of such data rests solely with participating institutions and Tribal governments.

Oversight will be provided by a NIHDPI Data Governance Board composed of AAIP, IHEART, and SGDI representatives. This board will review all DUAs, Tribal approvals, and proposed public releases; ensure alignment with FAIR data principles; and guarantee that Indigenous governance frameworks are upheld. Compliance documentation, including data management activities and governance decisions, will be reported through annual NSF progress reporting. Through this framework, NIHDPI ensures that all data are managed ethically, securely, and in accordance with both NSF expectations and Tribal sovereignty requirements.